

Session 02: Machine Learning vs Deep Learning and the End-to-End AI Workflow

This session establishes a clear conceptual understanding of the relationship between Machine Learning (ML), Deep Learning (DL), and Artificial Intelligence as a whole. It then introduces the standard AI workflow and contextualizes it within agricultural applications.

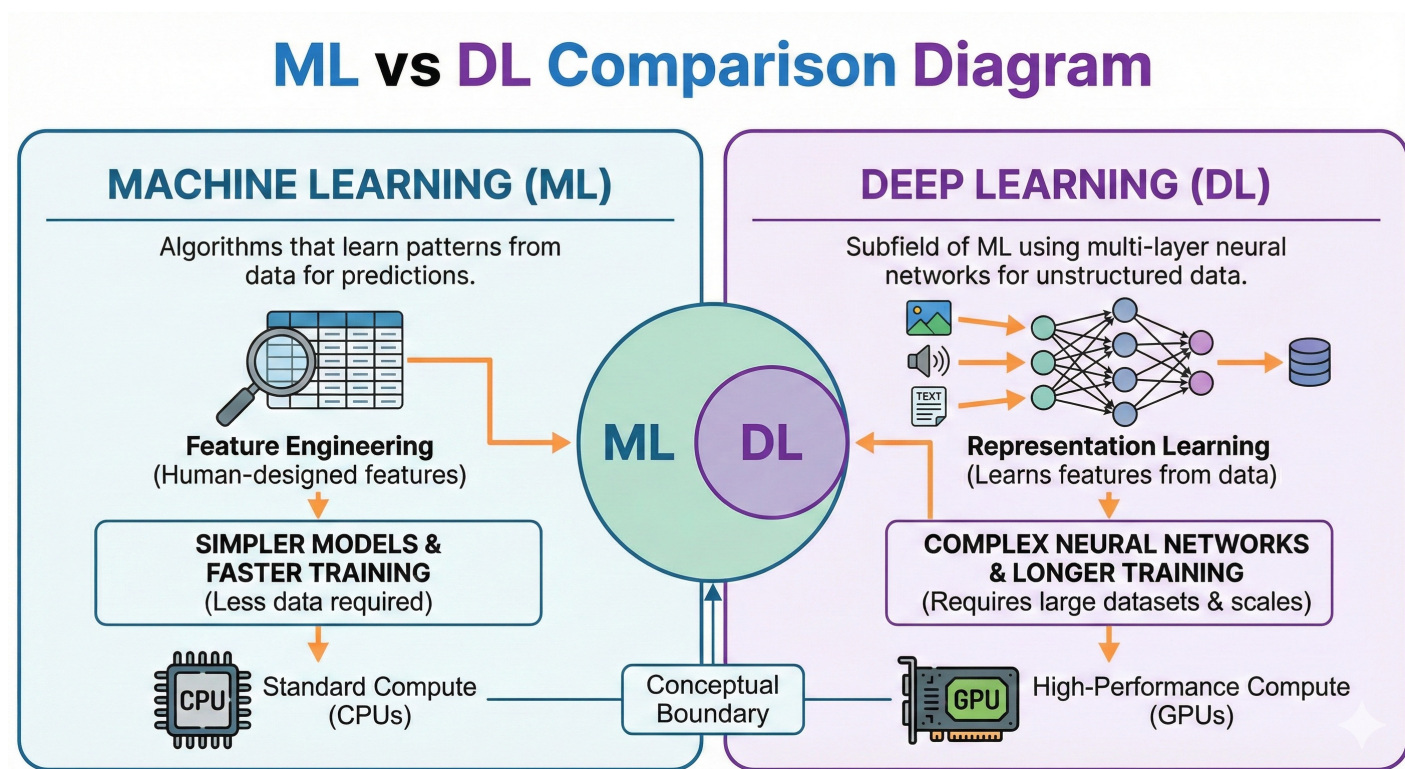
1. Conceptual Boundary Between ML and DL

Machine Learning encompasses algorithms that learn patterns from data to make predictions or decisions. These models typically rely on structured features engineered by humans.

Deep Learning is a specialized subfield of ML based on neural networks with multiple layers capable of learning hierarchical representations directly from data, particularly unstructured data such as images, audio, and text.

Key distinctions:

- ML relies more on feature engineering; DL relies on representation learning.
- ML models are often simpler, faster to train, and require less data.
- DL models scale better with large datasets and complex tasks.
- DL depends heavily on GPUs and high-performance compute environments.



2. Taxonomy of Machine Learning Paradigms

A structured map of ML categories.

Supervised Learning

Models learn from labeled data.

- Regression
- Classification
- Practical applications: yield prediction, disease detection

Unsupervised Learning

Models discover structure in unlabeled data.

- Clustering
- Dimensionality reduction
- Applications: soil type grouping, field segmentation

Reinforcement Learning

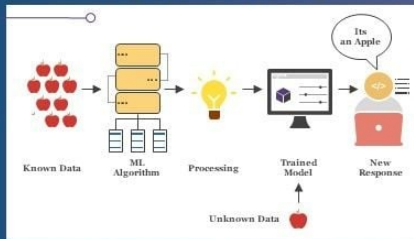
Agents learn by interacting with an environment and receiving rewards.

- Applications: autonomous drones, irrigation automation

Methodologies of ML: Supervised, unsupervised and reinforcement learning

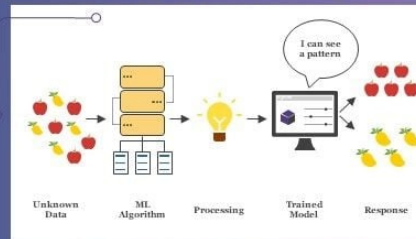
This slide provides information regarding major methodologies of machine learning approach such as supervised learning, unsupervised learning and reinforcement learning. In each methodology, training data is fed to the system for gaining relevant outcomes.

Supervised Learning



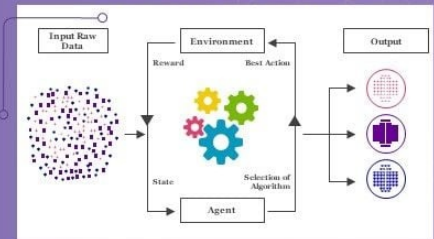
- o Utilize known or labeled data for training or supervised learning for successful execution
- o Input data utilize ML algorithm and is used for model training
- o Post model training by known data, unknown data is used into model to generate new response
- o Major algorithms used for supervised learning include
 - Polynomial regression
 - Linear regression
 - Random forest
 - Naive bayes
 - Logistic regression

Unsupervised Learning



- o Training data is unknown or unlabeled and input data is not guided to algorithm
- o Data is fed to ML algorithm and utilized to train model
- o Trained model focuses on detecting the pattern and offer desired outcome
- o Major algorithms used for unsupervised learning include
 - Partial least squares
 - K-means clustering
 - Fuzzy means
 - Hierarchical clustering
 - Principal component analysis

Reinforcement Learning



- o Feedback-based learning approach, which allows learning agent to get reward for each right actions and penalty for every wrong action
- o Agent learns with such feedback for performance enhancement
- o Agent interacts with environment and analyze it
- o Major components of RL include
 - Learning agent
 - Testing environment
 - Action

This slide is 100% editable. Adapt it to your need and capture your audience's attention.

3. Deep Learning Fundamentals

Core ideas underpinning modern DL systems.

Neural Networks

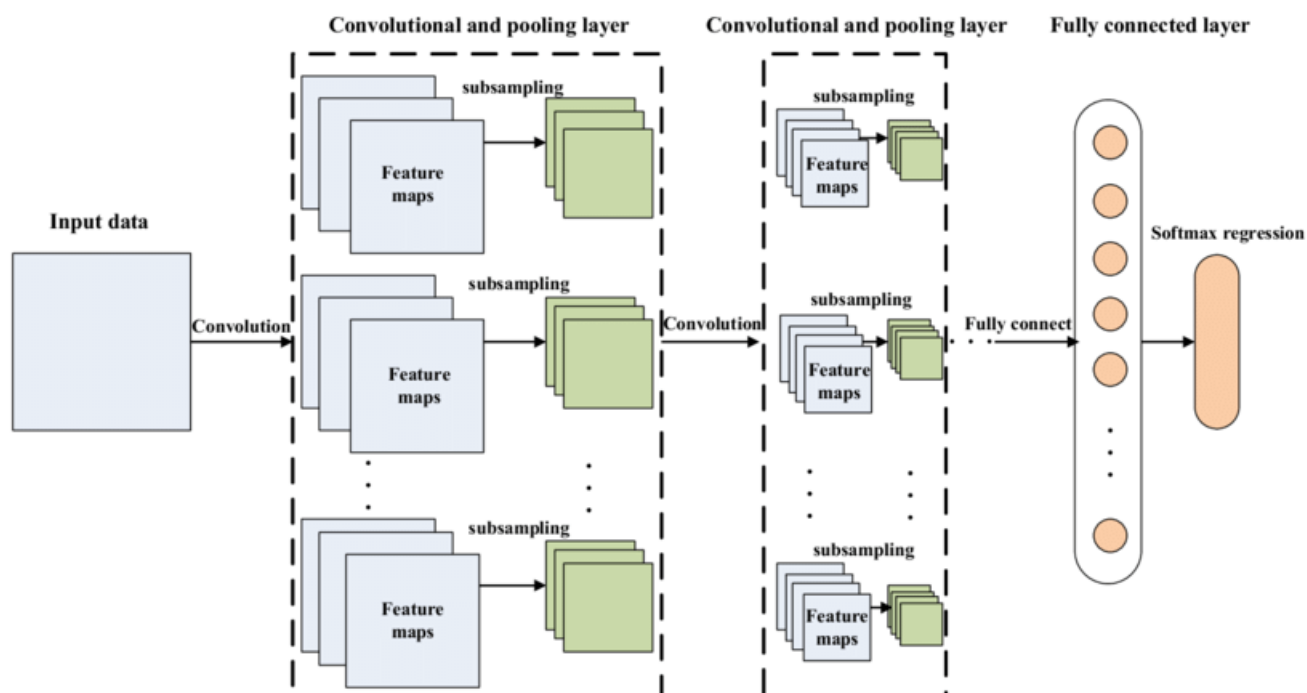
Layers of interconnected units (neurons) that transform inputs through learned weights.

Key Concepts

- Activation functions
- Loss functions
- Optimization (SGD, Adam)
- Training vs inference
- Overfitting and regularization

Common Architectures

- Convolutional Neural Networks (CNNs) for images
- Recurrent Neural Networks (RNNs) and LSTMs for sequences
- Transformers for general-purpose modeling

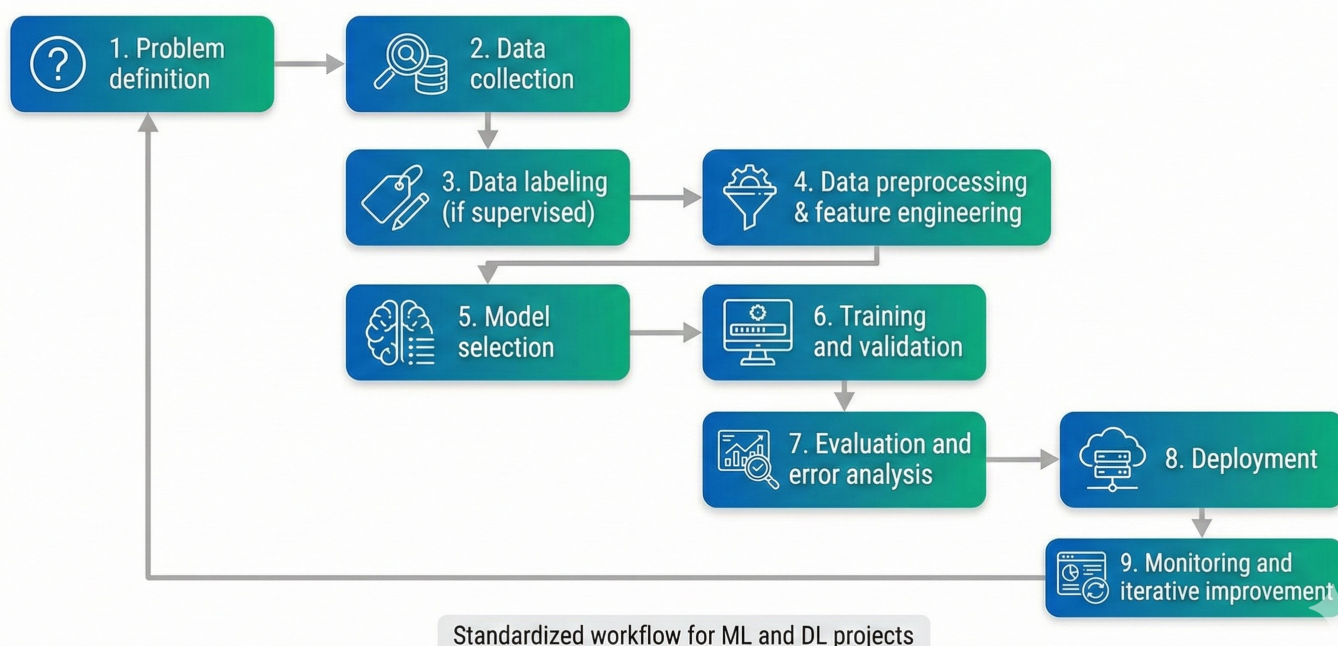


4. End-to-End AI Workflow

A standardized workflow used in both ML and DL projects.

1. Problem definition
2. Data collection
3. Data labeling (if supervised)
4. Data preprocessing and feature engineering
5. Model selection
6. Training and validation
7. Evaluation and error analysis
8. Deployment
9. Monitoring and iterative improvement

End-to-end AI Workflow Diagram



5. AI Workflow in Agriculture

Applying the general AI workflow to domain-specific agricultural problems.

Example Contexts

- Crop disease detection using images
- Yield forecasting using numerical and environmental data
- Soil classification using satellite data

Adaptations for Agriculture

- Data collection may involve drones, sensors, or satellite imagery.
- Labeling often requires agronomists or domain specialists.
- Environmental variability makes model generalization more difficult.
- Deployment may occur on edge devices in remote environments.

Challenge Exercise

Choose one agricultural use case (e.g., crop disease detection, soil classification, or yield prediction) and outline an end-to-end AI workflow for it. Your outline must include:

1. Problem definition
2. Data sources
3. Required labels (if any)
4. Preprocessing steps
5. Model type selection (ML or DL)
6. Evaluation criteria
7. Deployment considerations

The deliverable can be a Markdown document or a short slide deck.